

EXPERIMENT 8

AIM: Write a Program to implement Stop and Wait Protocol.

SOLUTION:

Algorithm: Stop-and-Wait ARQ

Step 1: Start.

Step 2: Read the total number of frames (**n**).

Step 3: Set the frame number **i = 0**.

Step 4: Send frame **i**.

Step 5: Wait for the acknowledgment (ACK).

Step 6: Check whether the ACK is received.

- If **ACK is received**, send the next frame (**i = i + 1**).
- Otherwise, a **timeout** occurs and retransmit the same frame.

Step 7: Repeat Steps 4 to 6 until all frames are transmitted successfully.

Step 8: Stop.

```
#include <stdio.h>
#include <conio.h>

void main()
{
    int n, i = 0;
    int ack;

    clrscr();

    printf("=====\n");
    printf("          STOP-AND-WAIT ARQ\n");
    printf("=====\n");

    printf("Enter the number of frames: ");
    scanf("%d", &n);

    while (i < n)
    {
        printf("\n-----\n");
        printf("Sender : Sending Frame %d\n", i);

        printf("Enter ACK for Frame %d (1=Yes, 0=No): ", i);
        scanf("%d", &ack);

        if (ack == 1)
        {
            printf("Receiver : Frame %d received.\n", i);
            printf("Receiver : ACK %d sent.\n", i);
            printf("Sender   : ACK %d received.\n", i);
            printf("Frame %d transmitted successfully.\n", i);
```

```

        i = i + 1;
    }
    else
    {
        printf("Receiver : Frame %d not received.\n", i);
        printf("Sender    : ACK not received.\n");
        printf("Timeout occurred.\n");
        printf("Retransmitting Frame %d...\n", i);
    }
}

printf("\n===== \n");
printf("All frames transmitted successfully.\n");
printf("===== \n");

getch();
}

```

Output: